Paper notes

One row per paper. Click a title to open the local copy in `papers/`. I drew the figures for these notes from each paper's text; they are not copies of the paper's own figures.
Charts with real numbers name the table they come from, and toy examples say so.

Paper	Method, explained with figures

01

Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models

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ECCV 2024 method name AnomalyRuler

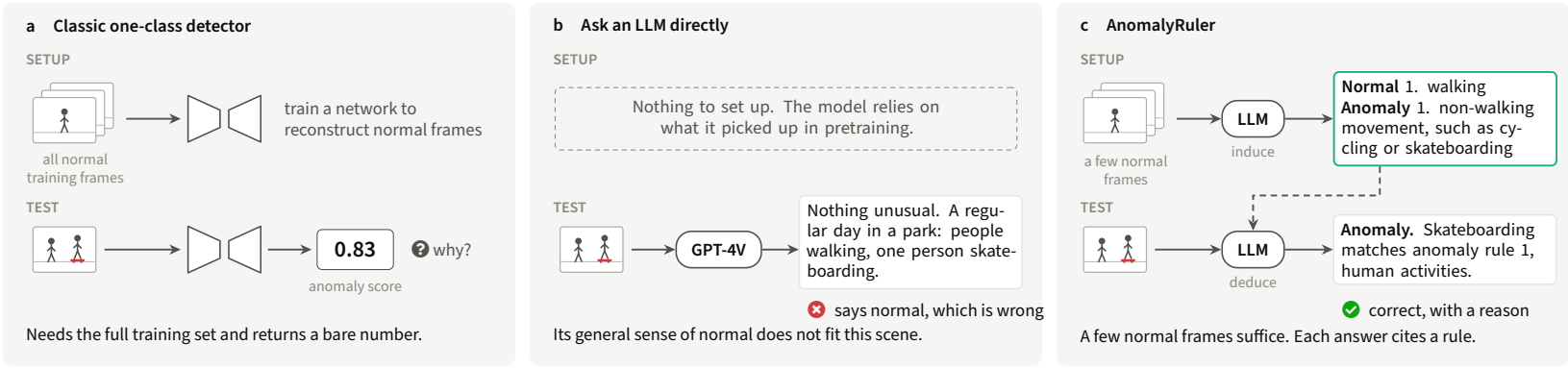
local copy
`papers/yang2024_eccv_follow_the_rules.pdf`
ECVA open-access PDF
ML Anthology page
code on GitHub
doi 10.1007/978-3-031-73004-7_18

one-class VAD · few normal shots · no training · LLM reasoning · explains each call

Why it is here. It explains each detection by naming a rule and needs only a few normal clips. Our brief asks for both.

1 The problem it fixes

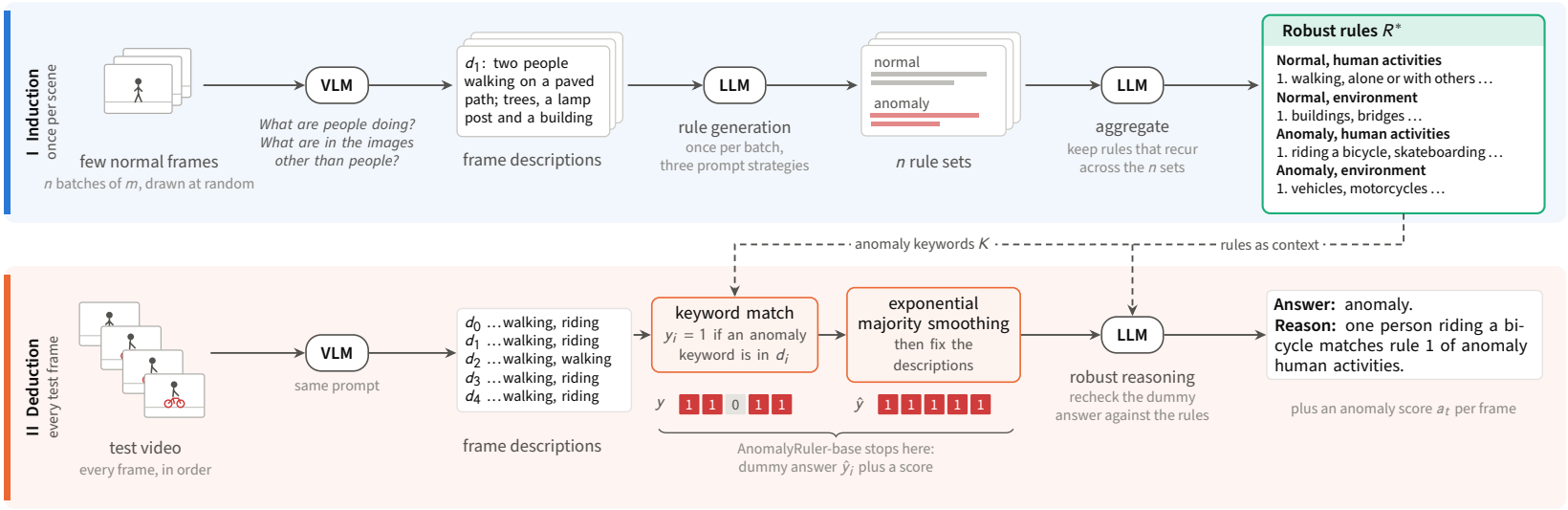
Scene: a campus walkway where only walking is allowed. The test frame shows one person walking and one skateboarding.



On this walkway only walking counts as normal, so the skateboarder is the anomaly. A classic detector trained on normal frames returns a high score and nothing else. An LLM asked straight out uses its general sense of normal, decides skateboarding is ordinary, and misses it. The paper's Fig. 1 shows GPT-4V making exactly this call.

AnomalyRuler writes down what normal means for this scene before it looks at any test frame. The rules come from a few normal frames, and every verdict cites the rule it matched. Nothing is trained, so a new scene needs new reference frames, not a new model.

2 The pipeline at a glance



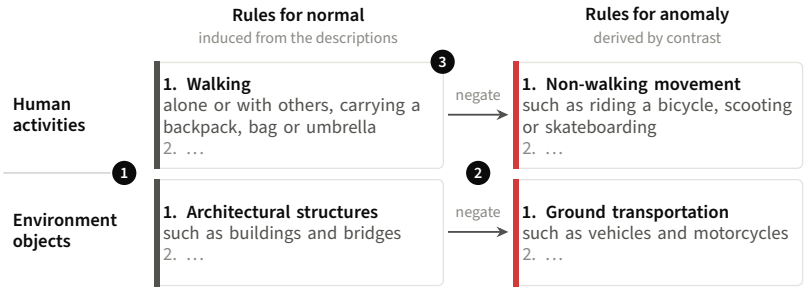
Induction runs once per scene. A VLM captions a few normal frames. An LLM reads each caption, writes rules for normal, then flips them into rules for anomaly. This repeats over n random batches, and one last LLM call keeps only the rules that keep coming back.

Deduction runs on every frame. The same VLM captions the test frame. Matching the caption against anomaly keywords gives a first 0 or 1. Smoothing over neighbouring frames cleans that guess, then an LLM rechecks it against the rules and writes the reason.

Paper setup. CogVLM-17B captions frames, GPT-4 writes the rules, and the open Mistral-7B runs deduction, since GPT-4 over whole test sets cost too much. Defaults are $n = 10$, $m = 1$, $p = 5$, $\alpha = 0.33$. All models stay frozen.

3 Induction: from normal frames to a rule book

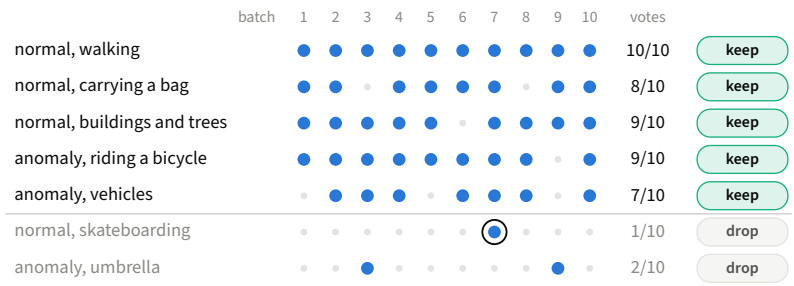
a Rule generation, three prompt strategies



1. Human and environment. The VLM prompt splits people from objects, so rules come in two families.
2. Normal and anomaly. Anomaly rules negate the normal ones, so no anomaly footage is needed.
3. Abstract and concrete. Each rule names a class, then examples, so new cases match by analogy.

Taking out the normal-to-anomaly contrast costs 18.8 AUC points on ShanghaiTech, the biggest drop in the ablation. With normal rules alone the LLM flags walking with an umbrella as a deviation and still misses real anomalies. Dropping the human and environment split halves the rule count, from 42 to about 22, and costs 2.4 points.

b Rule aggregation, keep what recurs



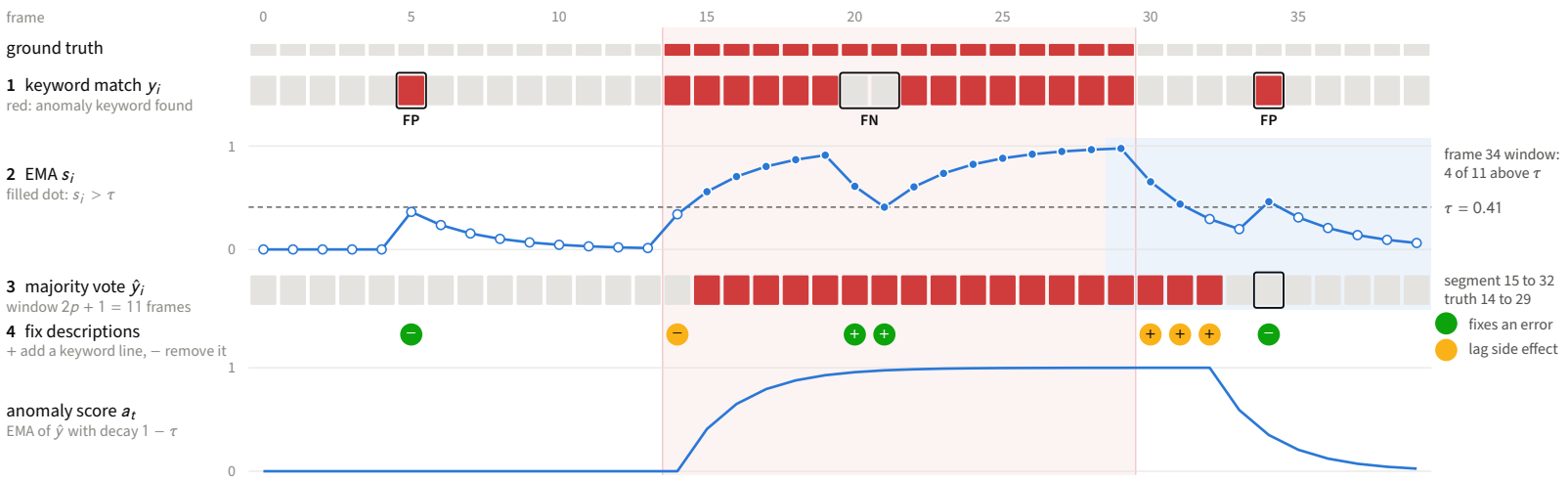
Batch 7: one caption misread a walker as a skateboarder, so a wrong normal rule shows up once and is voted out.
Illustrative data. In the paper an LLM runs this vote from a prompt over $n = 10$ rule sets.

This is randomized smoothing applied to prompts. One bad caption can plant a wrong rule, but the same mistake rarely repeats across ten independent batches. Without aggregation AUC falls 15.8 points. More reference frames help up to a point: at $n = 20$, AUC slides from 85.9 to 72.2 as m grows, because past about 100 frames the context gets too long.

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4 Deduction, step one: perception smoothing on a toy video



1 Keyword match. $y_i = 1$ if a keyword $k \in K$ taken from the anomaly rules appears in caption d_i, else 0. Keywords are mostly -ing verbs such as riding or running.

2 EMA. $s_t = \frac{\sum_{i=0}^{t}(1-\alpha)^{t-i}\, y_i}{\sum_{i=0}^{t}(1-\alpha)^{i}}$ with $\alpha = 0.33$. It only looks back in time.

3 Majority vote. $\hat{y}_i = 1$ when more than half of the s_j in the window $[i - p, i + p]$ exceed τ, the mean of all s. Default $p = 5$.

4 Fix and score. Where $y_i = 0$ but $\hat{y}_i = 1$, append *There is a person {k}.* Where $y_i = 1$ but $\hat{y}_i = 0$, delete the keyword phrase. The score a_t reruns the EMA on $\hat{y}$.

What the toy run shows. Forty frames, a true anomaly at frames 14 to 29, and the paper's defaults $\alpha = 0.33$ and $p = 5$. Every curve and cell above comes from the paper's formulas; only the input y is made up. Smoothing removes both lone false positives and fills the gap at frames 20 and 21. Frame 34 is the vote's job: its own EMA value clears τ, but only 4 of the 11 values in its window do.

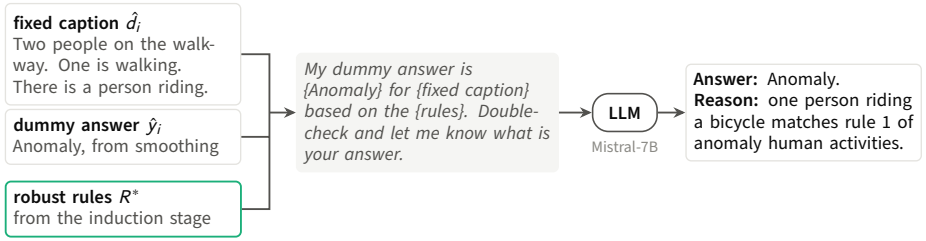
The cost is lag. The EMA only looks back, so the smoothed segment starts one frame late and ends three late. Frames 30 to 32 get a fake keyword line and frame 14 loses a true one. At 30 fps that is 0.1 s and does not matter. If we caption at 1 fps to fit the 5-minute budget it grows to 3 s, and the brief scores timecode accuracy.

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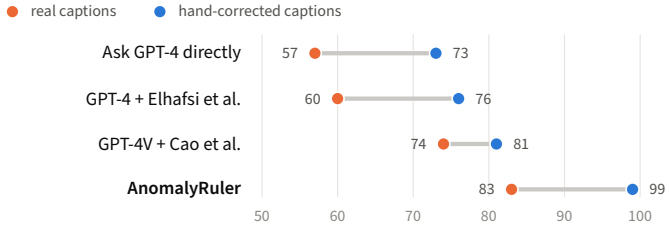
5 Deduction, step two: robust reasoning

a What the LLM sees in the recheck



b Right answer for the right reason, %

100 ShanghaiTech frames, doubly-right metric. Source: Table 2



The second LLM pass does not judge the frame from scratch. It gets the fixed caption, the dummy answer from smoothing and the rules, and is told to double-check. The paper reports that the hint cuts missed anomalies and ties each reason to a numbered rule. For AUC, Eq. 2 runs again on these final answers to give the score.

With hand-corrected captions AnomalyRuler is right for the right reason on 99 of 100 frames; with real captions, 83. Reasoning is not the weak link, perception is. Smoothing was off in this test because the frames are not consecutive.

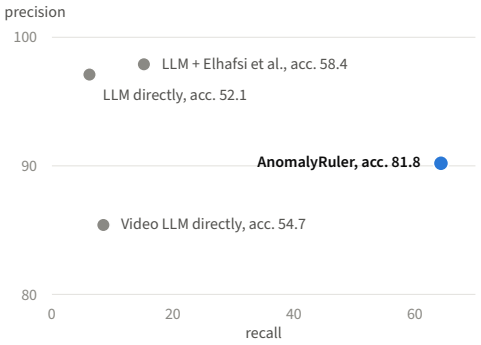
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6 Results

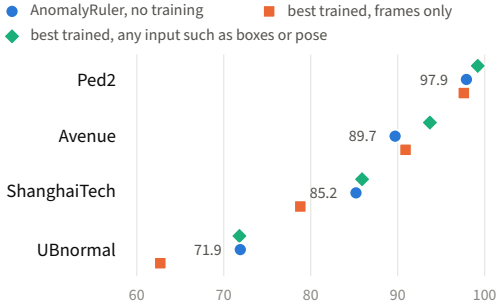
a Against other LLM pipelines

ShanghaiTech, %. Source: Table 1



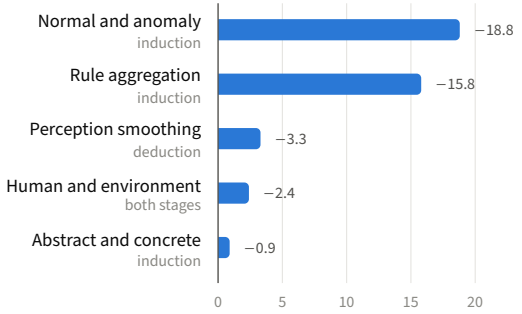
b Frame-level AUC against trained methods

%, best of 15 methods per group. Source: Table 3



c AUC lost when one part is removed

ShanghaiTech, full model 84.5. Source: Table 5



Asked directly, LLMs almost never say anomaly, so precision looks great and recall sits at 6 to 15%. AnomalyRuler finds 64% of anomalous frames at 90% precision.

With no training it beats every frames-only method on Ped2, ShanghaiTech and UBnormal, and trails the best one on Avenue by 1.2 points. Methods that add object boxes or pose still lead on Ped2, Avenue and ShanghaiTech.

Two parts carry the method: anomaly rules built by contrast, and the aggregation vote. The other three add 0.9 to 3.3 points each. Rules induced on ShanghaiTech and applied to Ped2 reach 97.4 AUC with the base version, against 95.7 for zxVAD, Table 4.

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7 What carries over to our ForteBank service

In the paper	In our service	
A few normal frames, sampled from a training set with no anomalies	Normal stretches of our 42 labelled videos, cut away from each violation timecode, plus clips we stage ourselves	change
CogVLM-17B captions each frame; the prompt splits people from everything else	A local VLM, with the prompt split by role: cashier, client or cash collector, then cash and items on the desk	change
GPT-4 writes the rules, Mistral-7B applies them	A local LLM for both stages. Section 7 of the brief bans sending frames or captions to any cloud AI	change
Rules come only from data	Each rule also carries its source: document and clause from NBRK acts on adilet.zan.kz. Induced rules describe what normal looks like	add
Output: a 0 or 1 per frame and an anomaly score	Smoothed segments become violations: timecode_sec at the segment start, 1 to 3 evidence frames near the score peak	change
A reason that names the matched rule	Goes straight into the description field of the API response	keep
Smoothing runs over every frame of the video	Caption at 1 to 2 fps to stay under 5 minutes per video, and take the start time from raw keyword hits because the EMA lags	change
Text rules transfer across datasets, Table 4	Should help with the new camera angle in the final test, which criterion 5 scores	keep

Desk-level detail. The paper's anomalies are people on bikes and cars in outdoor scenes. Ours are hands, cash and documents seen from one camera above the till. If the VLM cannot caption those, nothing downstream recovers.

Omissions. Keyword matching flags something that is there. A cashier who skips a required step leaves no keyword to match, so those rules need a check over a time window, not one frame.

One frame at a time. Apart from smoothing, each frame is judged alone. Actions that only make sense over a few seconds may need short clip captions instead.

False alarms. The bank accepts no misses and wants few false alarms. The recheck targets misses; the window p and threshold τ decide how many false alarms survive, so tune them on our 42 labelled videos.

Verdict. Worth a prototype. Test VLM captions on five training clips first, since everything depends on them. Then write the rule table with sources, then add smoothing and the recheck.